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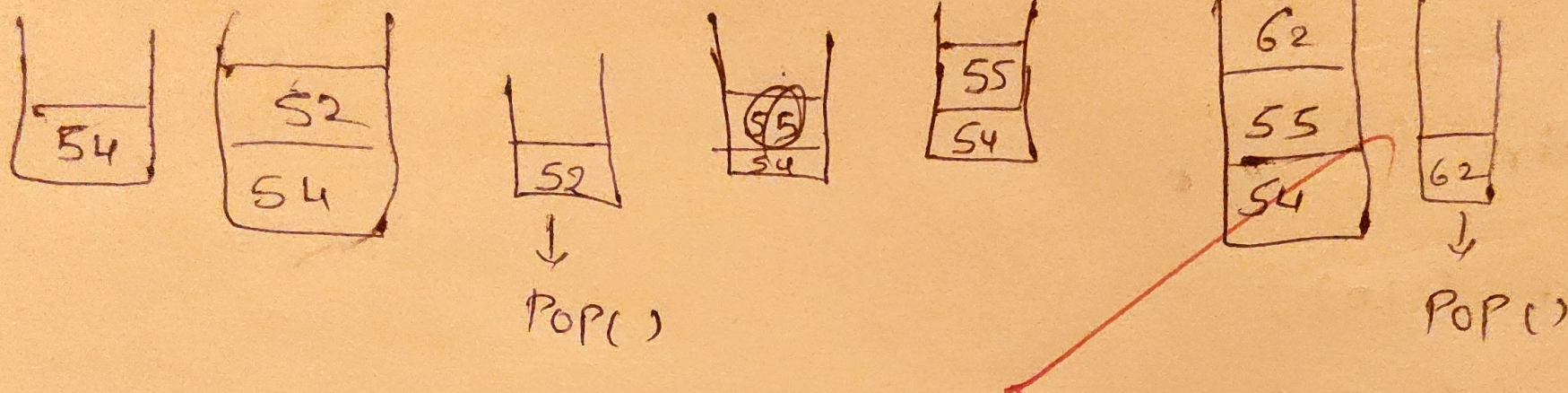
Activity :-

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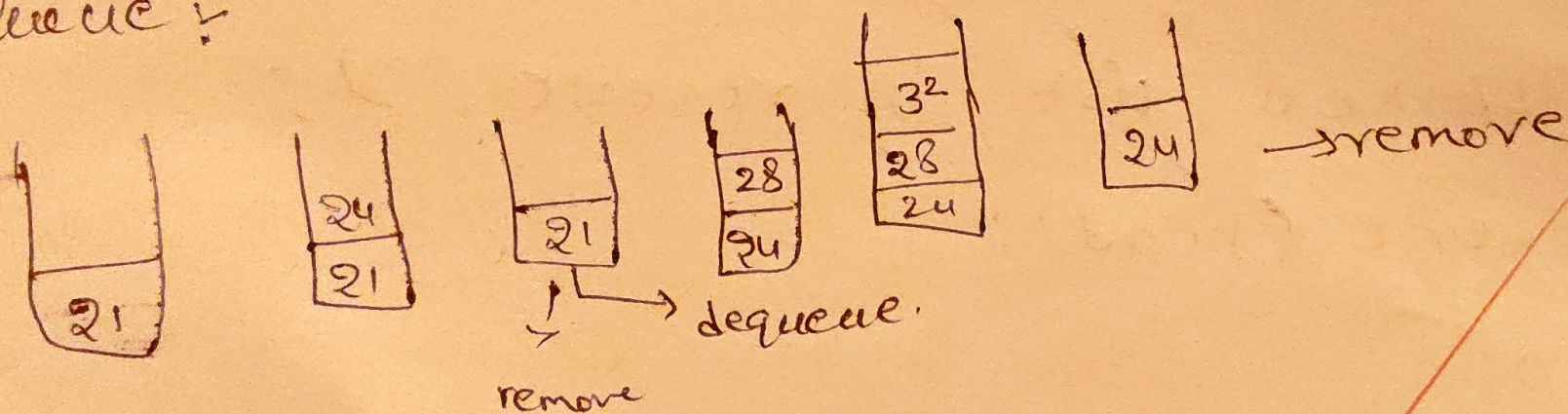
DSA0 C8A0311

1) Stack & Queue



Stack = 62

Queue :-



Queue = 24

ans :- $s + q = 62 + 24 = 86$

2) Prefix Postfix expression $10\ 5\ +\ 60\ 6\ / \ +\ 8\ -\ 15$

$$\left[\begin{array}{c} 5 \\ 10 \end{array} \right] + \left[\begin{array}{c} 60 \\ 15 \end{array} \right] \left[\begin{array}{c} 6 \\ 60 \\ 15 \end{array} \right] \div \left[\begin{array}{c} 10 \\ 15 \end{array} \right] + \left[\begin{array}{c} 8 \\ 150 \end{array} \right] - \left[\begin{array}{c} 150 \end{array} \right]$$

Ans: 142

3) Postfix expression: $8\ 2\ 3\ \wedge\ / \ 2\ 3\ * \ +\ 5\ 1\ * \ -$

Push 8: $\left[\begin{array}{c} 8 \end{array} \right]$ Push 2: $\left[\begin{array}{c} 2 \\ 8 \end{array} \right]$ Push 3: $\left[\begin{array}{c} 3 \\ 2 \\ 8 \end{array} \right]$

$$(2^3) = 8$$

$$\left[\begin{array}{c} 8 \\ 8 \end{array} \right] \cdot 8 \div 8 = 1$$

TOP = 6

Next = 1

answer: 6 and 1

$\left[\begin{array}{c} 1 \end{array} \right]$ $\left[\begin{array}{c} 2 \\ 1 \end{array} \right]$ $\left[\begin{array}{c} 3 \\ 2 \\ 1 \end{array} \right]$ $2 \times 3 = 6$

4) infix expression: $a + b \times c - d \wedge e \wedge f$

scan	Stack	Postfix
a		a
+	+	a
b		ab
*	+ *	ab
c	+ *	abc
-	-	abc + *
d	-	abc * + d
^	- ^	abc * + d
e	- ^	abc + + de
^	- ^ ^	abc * + de
f	- ^ ^	abc + + def
end		abc + def ^ ^ - *

5)

20
10

10
10

20
10
10

10
10

10

20
10

Pop Sequence : 20, 10, 10, 20

or